

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
ELECTRONICS AND COMMUNICATION ENGINEERING DEPARTMENT

Ops with java (IIPC-213)

Mid Sem. Exam – I

Date:

Max. Marks: 15

Duration: 50 min.

All questions are compulsory.

1. Create a base class Ticket with private members like ticketNumber and distance. Provide a method calculateFare(). Derive subclasses like TrainTicket, BusTicket, and FlightTicket. Each should have different fare calculation logic. Override the method and use runtime polymorphism.
2. Create a base class Consumer with private data members such as consumerID and unitsConsumed. Provide a virtual method calculateBill(). Derive subclasses such as DomesticConsumer and CommercialConsumer. Each subclass should override calculateBill() with different per-unit charges. Use runtime polymorphism to compute final electricity bill.
3. Create a base class Employee with private data members name and basicSalary. Derive Manager and Developer classes that override calculateSalary() method. Use runtime polymorphism to display final salary.